

CE 310 — Week 4 Weekly Quiz

Week: 4 | **Topic:** Discrete Events: The Poisson Distribution

Due: Closes **8:00 AM Tuesday, Sep 22** — the day before Exercise 4 and Homework 4 are due. Taken on D2L

Points: 15 questions $\times$ 1 pt each = 15 pts

Attempts: Two — the higher score counts

Time limit: 25 minutes

Instructions: Answer each question in the format it asks for — most are multiple choice, with one matching, one ordering, one select-all-that-apply and one true/false. This quiz is auto-graded and must be submitted before it closes at the time given above. You may use your notes and readings, but work independently.

This quiz covers the Monday lecture and the Wednesday Excel tutorial — the Poisson distribution, event rates, PMF vs. CDF, return periods, and the Excel functions that go with them. The tutorial's warning boxes are where the most time-consuming mistakes are flagged, so they are worth reading before you start.

Questions

Q1. In October 2015, a storm was labeled a “1,000-year flood” — yet similar events had occurred in the same city twice in the previous six years. The Poisson model reveals that:

- A) The three events confirm that 1,000-year floods are becoming more frequent due to climate change — no statistical error was made.
 - B) Three events in six years is consistent with a 1,000-year return period if the storms were clustered.
 - C) The Poisson model is inappropriate for flood events because rainfall is not discrete.
 - D) $P(X \geq 3 \text{ in } 6 \text{ years} \mid \lambda = 0.006)$ is essentially zero, which means the “1,000-year” label was almost certainly incorrect.
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Q2. A civil engineer says: “The 50-year storm will not happen during this 10-year project.” Why is this reasoning dangerous?

- A) Because the 50-year storm has already occurred within the last decade, resetting the clock.
- B) Because $P(\text{at least one 50-year storm in } 10 \text{ years}) = 1 - e^{-(10/50)} \approx 18\%$, which engineers cannot ignore.

- C) Because ASCE 7-22 requires designing for the 100-year storm, not the 50-year storm.
 - D) Because the engineer is confusing the return period with the recurrence interval.
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Q3. [Select all that apply] Which of the following situations are appropriately modelled by a Poisson distribution?

- (A) The number of equipment failures in a building per year, given a known average failure rate
 - (B) The compressive strength of a concrete cylinder
 - (C) The number of rain days recorded at a weather station in July
 - (D) Whether a single component passes or fails one inspection
 - (E) The number of vehicles arriving at an intersection in a five-minute interval
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Q4. [True/False] A structure designed for the 100-year storm has a 1% chance of being exceeded in any given year, and therefore roughly a 26% chance of being exceeded at least once during a 30-year service life.

Q5. Tucson's North American Monsoon produces most of the city's annual rainfall in July and August. The dataset records 103 rain days in July across 10 years. What is the correct July lambda (λ_{july})?

- A) $103 / 10 = 10.3$ rain days per July
 - B) $103 / 31 = 3.32$ rain days per July day
 - C) $103 / 120 = 0.858$ rain days per month
 - D) $103 / 30 = 3.43$ rain days per summer month
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Q6. Which statement correctly distinguishes the Poisson PMF from the Poisson CDF?

- A) The PMF gives the probability of exactly k events; the CDF gives the probability of k or fewer events.
 - B) The PMF gives the probability of at least k events; the CDF gives the probability of exactly k events.
 - C) The PMF is used for continuous variables; the CDF is used for discrete count variables.
 - D) The PMF and CDF are identical when lambda equals 1.
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Q7. For a Poisson random variable X with $\lambda = 3.86$, what is $E[X]$ (the expected value)?

- A) $\sqrt{3.86} \approx 1.96$
 - B) $3.86^2 = 14.90$
 - C) 3.86
 - D) Cannot be determined without the full PMF table.
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Q8. A facilities team estimates $\lambda = 2$ equipment faults per quarter. Using the Poisson PMF formula **by hand** (not Excel) and $e^{-2} \approx 0.1353$, what is $P(X = 1)$?

- A) 0.1353 — this is $P(X = 0)$, not $P(X = 1)$
 - B) 0.2706 — $P(X = 1) = e^{-2} \times 2^1 / 1! = 0.1353 \times 2$
 - C) 0.0677 — dividing by 1! as if it were 2!
 - D) 0.4055 — $e^{-2} \times 3$, an arithmetic slip
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Q9. [Matching] Match each Excel formula to the probability it returns for a Poisson variable with mean λ .

Choices:

- (A) =POISSON.DIST(k, λ, FALSE)
- (B) =COUNTIFS(range, ">0")
- (C) =POISSON.DIST(k, λ, TRUE)
- (D) =1 - POISSON.DIST(k, λ, TRUE)
- (E) =1 - POISSON.DIST(0, λ, TRUE)

Matches:

1. $P(X \leq k)$ — k events or fewer
 2. $P(X \geq 1)$ — at least one event
 3. $P(X = k)$ — exactly k events
 4. $P(X > k)$ — more than k events
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Q10. [Ordering] A drainage structure is being designed for a 10-year project life against a storm with a 50-year return period. Put the steps in the order you would carry them out to find the probability of at least one such storm during the project.

- (A) Convert the return period into an annual exceedance probability, $p = 1/50$
 - (B) Scale it to the project length to get the expected number of events, $\lambda = 10 \times p = 0.2$
 - (C) Compute the probability of no events over that span, $P(X = 0) = e^{-0.2}$
 - (D) Subtract from 1, giving $P(\text{at least one}) \approx 18\%$
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Q11. You count rows where a Storms column equals 1, and get 0 although the column is clearly full of 1s. What is the most likely cause?

- A) The criterion was typed as "1" in quotes, comparing text to numbers
 - B) COUNTIFS cannot test for equality at all, only greater-than and less-than
 - C) The column has to be sorted before a conditional count will read it
 - D) Excel requires COUNTIF rather than COUNTIFS for a single criterion
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Q12. For a Poisson model, λ is taken from the data directly. How?

- A) λ is the largest count observed in any period
 - B) λ is the total number of events divided by the number of periods observed
 - C) λ is the standard deviation of the counts
 - D) λ is fixed at 1 and the data is rescaled to match it
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Q13. =POISSON.DIST(2, 3.5, FALSE) and =POISSON.DIST(2, 3.5, TRUE) return different numbers. What is the difference?

- A) FALSE gives $P(X \leq 2)$ and TRUE gives $P(X = 2)$, the exact probability
 - B) FALSE returns the answer as a percentage while TRUE returns a plain decimal
 - C) FALSE gives $P(X = 2)$ exactly; TRUE gives $P(X \leq 2)$, the cumulative total
 - D) TRUE rounds the result to four decimal places and FALSE does not
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Q14. =POISSON.DIST(1, 2.4, FALSE) returns 0.2177. What do you enter where the probability is asked for as a percentage?

- A) 0.2177
 - B) 21.77
 - C) 2.177
 - D) 0.002177
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Q15. You need $P(\text{more than 2 storms})$ with $\lambda = 3.5$. Which formula is right?

- A) =1 - POISSON.DIST(2, 3.5, TRUE)
- B) =1 - POISSON.DIST(3, 3.5, TRUE)
- C) =POISSON.DIST(2, 3.5, TRUE)
- D) =1 - POISSON.DIST(2, 3.5, FALSE)